Team 292

M1: Approach — Green Finance Assistant

Megathon X: Singularity · PS1 — Green Customer Insights Platform (Lloyds Banking Group)

1. Problem statement and our understanding

Retail banks sit on rich transaction data but have no scalable, trustworthy way to turn it into green-finance action. PS1 asks for an AI platform that analyses customer transactions, products and behaviour to: (a) detect opportunities for green financial products — EV finance, green home loans, solar finance and paperless banking (plus optional green investments and green loans); (b) estimate the potential sustainability impact of each opportunity; (c) generate personalised sustainability insights for customers; (d) give business and relationship teams portfolio-level, drill-down insight; (e) encourage paperless and digital journeys; and (f) explain why every opportunity was identified.

Our understanding of the decisive constraint: **trust and traceability matter more than model complexity.** Every recommendation must carry evidence, confidence, reason codes and a data timestamp; products a customer is ineligible for must never be recommended; and impact figures must be reproducible estimates, not model outputs. We therefore enforce strict separation of concerns: **ML predicts, a deterministic carbon engine calculates, the LLM only explains, and rule-based checks decide eligibility.**

2. Idea / abstract

Green Finance Assistant is an explainable green customer intelligence platform for retail banking in India. An interpretable weighted-score model computes propensity for each customer × product pair and decomposes every score into human-readable reason codes backed by transaction evidence. A deterministic carbon engine converts opportunities into kg CO2e per year using documented emission factors (CEA India grid baseline 0.82 kg/kWh, IPCC 2006 fuel and gas factors, DEFRA 2023 paper factors), exposing the exact formula and assumptions behind every figure. A strictly grounded LLM layer narrates these structured outputs in plain English for customers and relationship managers, with a deterministic template fallback so the system can never hallucinate offline. A customer-memory layer records responses (interested / not for me) and re-orders future recommendations without ever mutating propensity, eligibility or impact. The platform ships as two experiences: a **Customer Green Companion** (personal scorecard, opportunities, grounded chat) and a **Lloyds Intelligence Console** (portfolio KPIs, segments, searchable customers, drill-down to a single transaction).

3. High-level approach, pipeline and architecture

Data. Synthetic Indian retail dataset (fixed seed 42, fully documented generation): 120 fictional customers, ~13,100 transactions over a trailing 12-month window, each assigned to exactly one of eight mutually exclusive sustainability categories (Fuel, Electricity, Gas, Public Transport, Home Improvement, Automotive, Utilities, Other). No real or confidential banking data is used. Injected duplicates exercise the data-quality path.

Pipeline.

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transactions → validation & duplicate flagging → categorisation → customer feature store (12-month windows) → interpretable propensity model (documented weights, sigmoid squashing) → deterministic eligibility rules → carbon engine (factor registry + fixed formulas) → recommendation store → LLM explanation layer (grounded, streaming) → dashboards & API
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- **Propensity model:** transparent weighted scoring per product; every weight documented; scores decompose into reason codes (e.g. HIGH_FUEL_SPEND, AGING_ICE_VEHICLE) with evidence strings.
- **Eligibility:** separate deterministic rules (age, income band, home ownership, existing EV); ineligible products are recorded for audit but never recommended.
- **Carbon engine:** rules-only module; same code path powers customer estimates and an interactive calculator; outputs labelled 'estimated potential impact'.
- **LLM layer:** receives only structured model/engine outputs; forbidden from predicting, calculating or implying eligibility; streams explanations and grounded chat answers; deterministic fallback on any failure.
- **Stack:** FastAPI + MongoDB backend (REST + SSE streaming), React + Tailwind + Recharts frontend, LLM via API. Reproducible from a fixed seed.

4. Planned testing methodology

- **Data quality:** schema validation, duplicate detection rate, categorisation coverage (target 100%), 12-month window completeness.
- **Model evaluation:** synthetic take-up labels from a documented, independent noisy rule conditioned on eligibility; customer-level train/validation/test splits via sha256(customer_id) hashing (60/20/20, disjoint — no leakage).

Metrics: AUC, precision@k, Brier score, log-loss, reported per product and macro-averaged against random and single-signal baselines, with 95% seeded bootstrap confidence intervals so uncertainty at this sample size is explicit.

- **Carbon engine:** known-answer unit tests per scenario (EV, solar, home retrofit, paperless) — expected vs actual, run on every evaluation pass.
- **Guardrail tests (live):** ineligible products are never recommended; every recommendation carries evidence, confidence, reason codes, timestamp and an impact estimate; memory never alters scores, eligibility or impact.
- **LLM checks:** fallback determinism (identical context → identical narration), number-grounding (every figure in the fallback narration must appear in the structured context), and read-only architecture over stored outputs.
- **Reproducibility:** fixed seed; dataset regeneration checksum (customers, transactions and amount totals must match the stored dataset).
- **Scenario tests:** missing data, ineligible customers and LLM-outage paths exercised via the fallback and guardrail harness.

Known limitations (to be reported): synthetic labels measure ranking behaviour against a documented rule, not real-world take-up; a 120-customer portfolio limits statistical power (surfaced through bootstrap confidence intervals); impact estimates are gross of rebound effects and behavioural change.